

ADCA – ADVANCED DIPLOMA IN COMPUTER APPLICATIONS

SHORT COURSE SYLLABUS

Duration: 12 Months | Level: Beginner to Advanced | Mode: Theory + Practical + Projects

Course Objective: To develop complete computer, office, accounting, programming, internet and job-oriented practical skills.

SEMESTER 1 – COMPUTER & OFFICE APPLICATIONS

- 1. Computer Fundamentals & Windows:** Computer basics, hardware/software, input-output devices, memory, CPU, Windows, files/folders, settings, shortcuts and basic troubleshooting.
- 2. Typing Skills:** English & Hindi typing practice, keyboard techniques, speed and accuracy improvement.
- 3. Microsoft Word:** Document creation, formatting, tables, page setup, mail merge, resume, letters, notices, certificates, reports and PDF export.
- 4. Microsoft Excel:** Worksheets, formulas, functions, IF, SUMIF, COUNTIF, lookup functions, sorting, filtering, charts, Pivot Table, data validation and dashboards.
- 5. Microsoft PowerPoint:** Professional presentations, themes, images, SmartArt, charts, transitions, animations, slide show and presentation design.
- 6. Internet & Digital Literacy:** Web browsing, search techniques, Gmail, attachments, Google Drive, Docs, Sheets, Forms, Meet, online forms and digital services.
- 7. Cyber Security:** Safe browsing, strong passwords, phishing, malware, OTP/UPI fraud, privacy, backup and cyber awareness.

SEMESTER 1 PRACTICAL WORK

- Resume, application, notice, certificate and project report in MS Word.
- Student marksheet, attendance, salary, billing and data-analysis sheets in MS Excel.
- Professional PowerPoint presentation with charts, images, animation and slide show.
- Email, online form, document upload/download, Google Forms and cloud storage practice.

MONTH-WISE PLAN – FIRST 6 MONTHS

Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Basics + Windows	MS Word	MS Excel	Advanced Excel + PPT	Internet + Cyber	Tally Basics

SEMESTER 2 – ADVANCED & JOB-ORIENTED SKILLS

8. Tally Prime with GST: Accounting fundamentals, company creation, ledgers, vouchers, sales, purchase, inventory, GST, payroll and financial reports.

9. DBMS & MySQL: Database concepts, tables, keys, relationships and SQL commands: CREATE, INSERT, SELECT, UPDATE, DELETE, WHERE, GROUP BY, HAVING and JOIN.

10. C Programming: Programming basics, variables, operators, conditions, loops, arrays, strings, functions, structures and practical programs.

11. HTML, CSS & JavaScript: Web page structure, forms, tables, styling, responsive design, basic JavaScript, DOM and interactive mini projects.

12. Graphics & Digital Design: Canva/basic photo editing, posters, certificates, ID cards, visiting cards, thumbnails and social-media designs.

13. Hardware & Networking: Computer components, Windows installation basics, drivers, bootable USB, printer/scanner, LAN/WAN, Wi-Fi, IP address and troubleshooting.

14. AI & Modern Digital Tools: Generative AI, prompt writing, document support, presentations, Excel/programming assistance, research, privacy and responsible AI use.

15. Employability & Office Skills: CV, professional email, interview preparation, office documentation, data entry, job portals and freelancing basics.

FINAL PRACTICAL & PROJECTS

- Tally business accounting project with GST entries and reports.
- Student/Employee database using MySQL and SQL queries.
- 40+ basic C programming exercises and problem-solving practice.
- Responsive personal portfolio or institute website using HTML, CSS and JavaScript.
- Final major project: Student Management, Library, Billing, Institute or Business Website.

MONTH-WISE PLAN – LAST 6 MONTHS

Month 7	Month 8	Month 9	Month 10	Month 11	Month 12
Tally + GST	DBMS + SQL	C Programming	HTML + CSS	JS + Design + AI	Hardware + Project

ASSESSMENT PATTERN

Monthly Tests	Assignments	Practical	Final Project	Viva / Exam
10%	10%	30%	20%	30%

COURSE OUTCOME

- Operate computers confidently and perform professional office work.
- Create documents, spreadsheets, presentations, accounts and basic databases.
- Use internet, cloud services, AI tools and digital platforms safely.
- Develop basic programs and websites and troubleshoot common computer problems.
- Prepare for roles such as Computer Operator, Data Entry Operator, Office Assistant and other entry-level digital jobs.